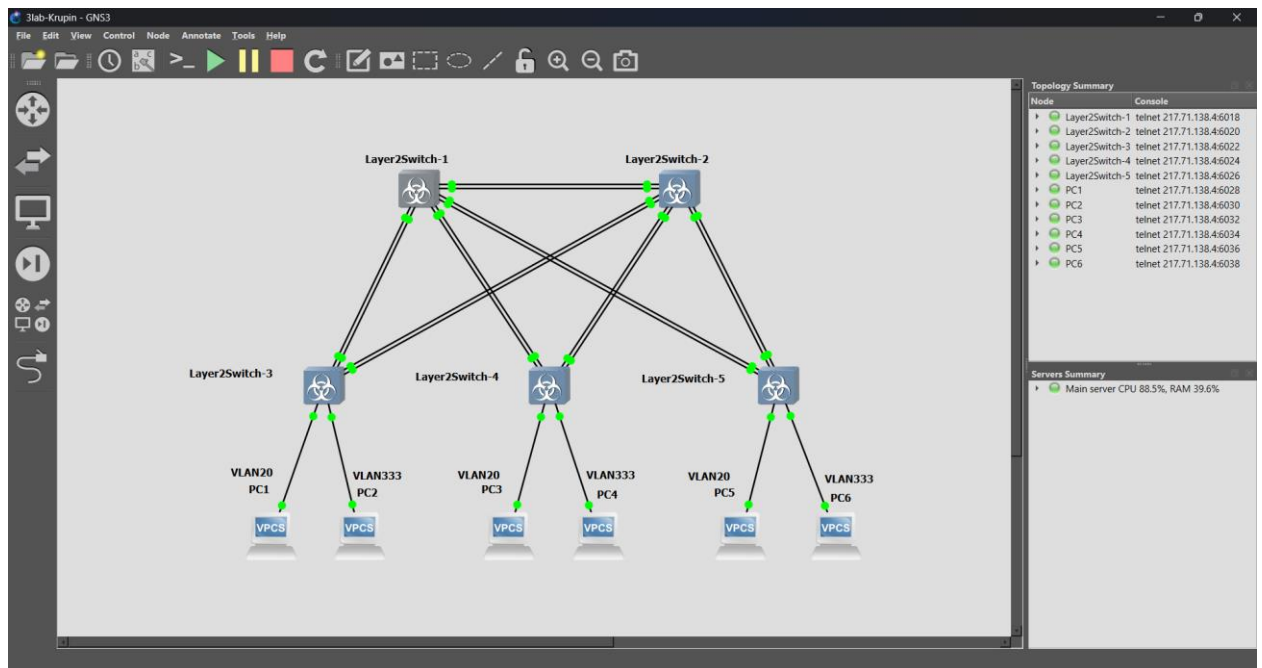




На каждом коммутаторе выполняем
show cdp neighbors

Layer2Switch-1

enable

conf t

vlan 20

name VLAN20

vlan 333

name VLAN333

interface range gi0/0-3,gi1/0-3

switchport trunk encapsulation dot1q

switchport mode trunk

switchport trunk native vlan 333

switchport trunk allowed vlan 20,333

end

write memory

show interfaces trunk

Layer2Switch-2

enable

conf t

vlan 20

name VLAN20

vlan 333

name VLAN333

interface range gi0/0-3,gi1/0-3

switchport trunk encapsulation dot1q

switchport mode trunk

switchport trunk native vlan 333

switchport trunk allowed vlan 20,333

end

write memory

Layer2Switch-3

enable

conf t

vlan 20

name VLAN20

vlan 333

name VLAN333

interface range gi0/0-3

switchport trunk encapsulation dot1q

switchport mode trunk

switchport trunk native vlan 333

switchport trunk allowed vlan 20,333

interface gi1/0

switchport mode access

switchport access vlan 20

interface gi1/1

switchport mode access

switchport access vlan 333

end

write memory

Layer2Switch-4

enable

conf t

vlan 20

name VLAN20

vlan 333

name VLAN333

interface range gi0/0-3

switchport trunk encapsulation dot1q

switchport mode trunk

switchport trunk native vlan 333

switchport trunk allowed vlan 20,333

interface gi1/0

switchport mode access

switchport access vlan 20

interface gi1/1

switchport mode access

switchport access vlan 333

end

write memory

Layer2Switch-5

enable

conf t

vlan 20

name VLAN20

vlan 333

name VLAN333

interface range gi0/0-3

switchport trunk encapsulation dot1q

switchport mode trunk

switchport trunk native vlan 333

switchport trunk allowed vlan 20,333

interface gi1/0

switchport mode access

switchport access vlan 20

interface gi1/1

switchport mode access

switchport access vlan 333

end

write memory

Назначаем IP-адреса

VLAN20

PC1

ip 192.168.20.1/24

PC3

ip 192.168.20.2/24

PC5

ip 192.168.20.3/24

VLAN333

PC2

ip 192.168.33.1/24

PC4

ip 192.168.33.2/24

PC6

ip 192.168.33.3/24

Проверяем

На PC1:

ping 192.168.20.2

ping 192.168.20.3

На PC2:

ping 192.168.33.2

ping 192.168.33.3

Не должно работать

На PC1:

ping 192.168.33.1

```
PC1 - PuTTY

PC1> ping 192.168.20.2

84 bytes from 192.168.20.2 icmp_seq=1 ttl=64 time=20.620 ms
84 bytes from 192.168.20.2 icmp_seq=2 ttl=64 time=23.917 ms
84 bytes from 192.168.20.2 icmp_seq=3 ttl=64 time=18.457 ms
84 bytes from 192.168.20.2 icmp_seq=4 ttl=64 time=17.894 ms
84 bytes from 192.168.20.2 icmp_seq=5 ttl=64 time=16.023 ms

PC1> ping 192.168.20.3

84 bytes from 192.168.20.3 icmp_seq=1 ttl=64 time=22.405 ms
84 bytes from 192.168.20.3 icmp_seq=2 ttl=64 time=16.927 ms
84 bytes from 192.168.20.3 icmp_seq=3 ttl=64 time=18.682 ms
84 bytes from 192.168.20.3 icmp_seq=4 ttl=64 time=17.679 ms
84 bytes from 192.168.20.3 icmp_seq=5 ttl=64 time=30.508 ms

PC1> ping 192.168.33.1

No gateway found

PC1> 
```

На PC2:

ping 192.168.20.1

```
PC2 - PuTTY

No gateway found

PC2> ping 192.168.33.2

84 bytes from 192.168.33.2 icmp_seq=1 ttl=64 time=21.554 ms
84 bytes from 192.168.33.2 icmp_seq=2 ttl=64 time=17.285 ms
84 bytes from 192.168.33.2 icmp_seq=3 ttl=64 time=24.623 ms
84 bytes from 192.168.33.2 icmp_seq=4 ttl=64 time=22.845 ms
84 bytes from 192.168.33.2 icmp_seq=5 ttl=64 time=18.322 ms

PC2> ping 192.168.33.3

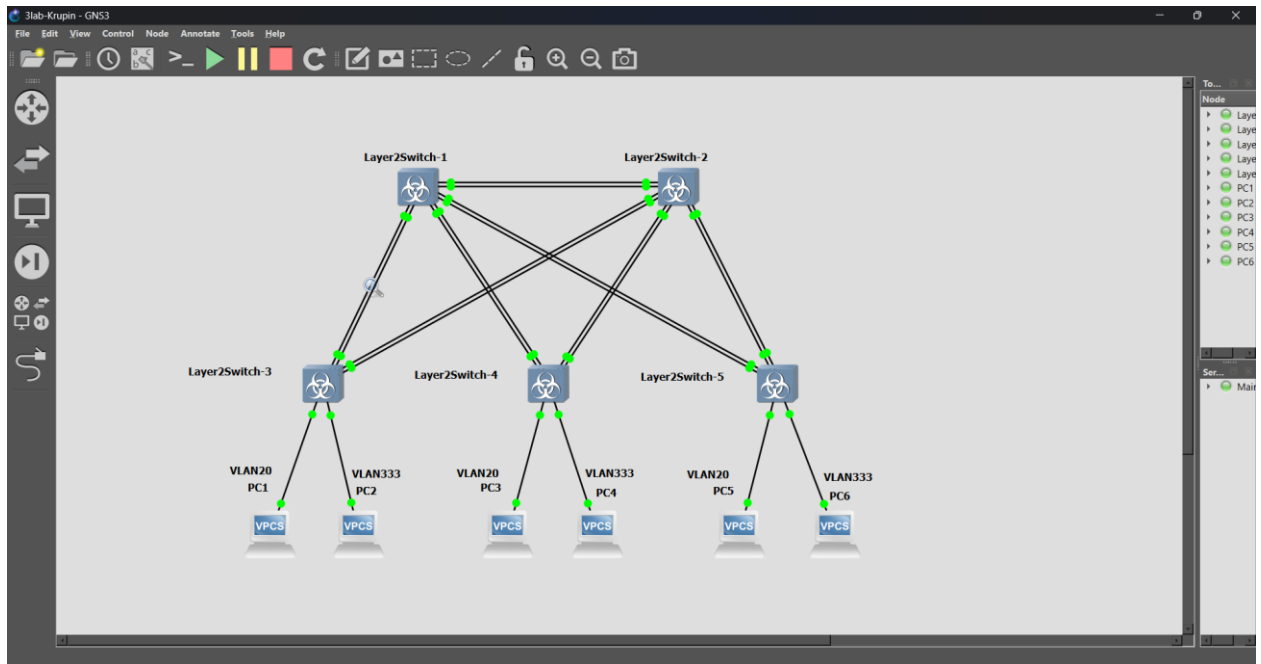
84 bytes from 192.168.33.3 icmp_seq=1 ttl=64 time=9.117 ms
84 bytes from 192.168.33.3 icmp_seq=2 ttl=64 time=18.977 ms
84 bytes from 192.168.33.3 icmp_seq=3 ttl=64 time=22.592 ms
84 bytes from 192.168.33.3 icmp_seq=4 ttl=64 time=24.247 ms
84 bytes from 192.168.33.3 icmp_seq=5 ttl=64 time=45.775 ms

PC2> ping 192.168.20.1

No gateway found

PC2> 
```

Перехватить пакеты в Wireshark



На PC1:

ping 192.168.20.2

No.	Time	Delta	Source	Destination	Protocol	Length	Info
3	0.145477	0.077408	0a:6a:1b:53:00:00	0a:6a:1b:53:00:00	LOOP	60	Reply
4	1.013630	1.748159	0a:6a:1b:53:00:00	0a:6a:1b:53:00:00	LOOP	60	Reply
5	2.350786	0.437156	0a:6a:1b:53:00:00	0a:6a:1b:53:00:00	LOOP	60	Reply
6	2.403103	0.052317	0a:6a:1b:53:00:00	0a:6a:1b:53:00:00	LOOP	60	Reply
7	4.585808	2.182705	0a:6a:1b:53:00:00	0a:6a:1b:53:00:00	LOOP	60	Reply
8	4.657834	0.072026	0a:6a:1b:53:00:00	0a:6a:1b:53:00:00	LOOP	60	Reply
9	6.953591	2.296057	0a:6a:1b:53:00:00	0a:6a:1b:53:00:00	LOOP	60	Reply
10	7.026513	0.072922	0a:6a:1b:53:00:00	0a:6a:1b:53:00:00	LOOP	60	Reply
11	8.480256	1.453743	Private_66:68:02	Private_66:68:02	Broadcast	ARP	68 Who has 192.168.20.2? Tell 192.168.20.1
12	8.491633	0.011377	Private_66:68:02	Private_66:68:02	Broadcast	ARP	68 192.168.20.2 is at 00:50:79:66:68:02
13	8.500051	0.008418	192.168.20.1	192.168.20.2	ICMP	102	Echo (ping) request id=0x6926, seq=1/256, ttl=64 (request in 14)
14	8.515588	0.014635	192.168.20.2	192.168.20.1	ICMP	102	Echo (ping) reply id=0x6926, seq=1/256, ttl=64 (reply in 13)
15	9.363203	0.847613	0a:6a:1b:53:00:00	0a:6a:1b:53:00:00	LOOP	60	Reply
16	9.418205	0.055002	0a:6a:1b:53:00:00	0a:6a:1b:53:00:00	LOOP	60	Reply
17	9.521603	0.103398	192.168.20.1	192.168.20.2	ICMP	102	Echo (ping) request id=0x6926, seq=2/512, ttl=64 (request in 18)
18	9.545592	0.024899	192.168.20.2	192.168.20.1	ICMP	102	Echo (ping) reply id=0x6926, seq=2/512, ttl=64 (reply in 17)
19	9.845119	0.299227	0a:6a:1b:53:00:00	0a:6a:1b:53:00:00	LOOP	60	Reply
20	10.548959	0.703870	192.168.20.1	192.168.20.2	ICMP	102	Echo (ping) request id=0x6926, seq=3/768, ttl=64 (request in 21)
21	10.558348	0.009359	192.168.20.2	192.168.20.1	ICMP	102	Echo (ping) reply id=0x6926, seq=3/768, ttl=64 (reply in 20)
22	11.575716	1.017368	192.168.20.1	192.168.20.2	ICMP	102	Echo (ping) request id=0x6926, seq=4/1024, ttl=64 (request in 23)
23	11.598080	0.022364	192.168.20.2	192.168.20.1	ICMP	102	Echo (ping) reply id=0x6926, seq=4/1024, ttl=64 (reply in 22)
24	11.758083	0.160003	0a:6a:1b:53:00:00	0a:6a:1b:53:00:00	LOOP	60	Reply
25	11.814677	0.056594	0a:6a:1b:53:00:00	0a:6a:1b:53:00:00	LOOP	60	Reply
26	12.350712	0.536035	0a:6a:1b:53:00:00	0a:6a:1b:53:00:00	LOOP	60	Reply

> Frame 13: Packet, 102 bytes on wire (816 bits), 102 bytes captured (816 bits) on interface -, id 0

> Ethernet II, Src: Private_66:68:02 (00:50:79:66:68:02), Dst: Private_66:68:02 (00:50:79:66:68:02)

> Destination: Private_66:68:02 (00:50:79:66:68:02)

> Source: Private_66:68:02 (00:50:79:66:68:02)

> Type: 802.1Q Virtual LAN (0x8100)

[Stream index: 4]

> 802.1Q Virtual LAN, Prio: 0, Dst: 0, Id: 20

> Internet Protocol Version 4, Src: 192.168.20.1, Dst: 192.168.20.2

> Internet Control Message Protocol

0000 00 50 79 66 68 02 00 50 79 66 68 00 81 00 00 14 Pyth F yfh
0010 00 00 45 00 00 14 26 01 00 00 40 01 aa ee c0 ad E Tg 8
0020 14 01 c0 a8 14 02 08 00 b8 e4 67 26 00 01 08 09g.....
0030 0a 0b 0c 0d 0e 0f 10 11 12 13 14 15 16 17 18 19
0040 1a 1b 1c 1d 1e 1f 20 21 22 23 24 25 26 27 28 29
0050 2a 2b 2c 2d 2e 2f 30 31 32 33 34 35 36 37 38 39
0060 3a 3b 3c 3d 3e 3f

На PC2:

ping 192.168.33.2

The image shows a Wireshark packet capture of an ICMP Echo (ping) session. The packet list on the left shows several ICMP Echo requests and replies. The packet details pane on the right shows the structure of the captured packet, including the Ethernet II header, Internet Protocol Version 4 header, and Internet Control Message Protocol header. The packet bytes pane on the right shows the raw data of the packet in hexadecimal and ASCII.

No.	Time	Delta	Source	Destination	Protocol	Length	Info
41	24.124947	0.330702	0c:6a:1b:53:00:00	0c:6a:1b:53:00:00	LOOP	60	Reply
42	28.177812	2.032865	0c:12:a7:54:00:00	192.168.33.2	ICMP	68	Conf. Root = 32768/33/0c:12:a7:54:00:00 Cost = 4 Port = 0x8003
43	28.229349	0.051437	0c:12:a7:54:00:00	192.168.33.2	ICMP	68	Conf. Root = 32768/33/0c:12:a7:54:00:00 Cost = 4 Port = 0x8003
44	28.518171	2.288822	0c:12:a7:54:00:00	192.168.33.2	ICMP	68	Conf. Root = 32768/33/0c:12:a7:54:00:00 Cost = 4 Port = 0x8003
45	29.589355	0.071184	0c:12:a7:54:00:00	192.168.33.2	ICMP	68	Conf. Root = 32768/33/0c:12:a7:54:00:00 Cost = 4 Port = 0x8003
46	29.538887	0.749532	0c:12:a7:54:00:00	192.168.33.2	ICMP	62	Dynamic Trunk Protocol
47	30.932072	1.593185	0c:12:a7:54:00:00	192.168.33.2	ICMP	68	Conf. Root = 32768/33/0c:12:a7:54:00:00 Cost = 4 Port = 0x8003
48	30.999339	0.067267	0c:12:a7:54:00:00	192.168.33.2	ICMP	68	Conf. Root = 32768/33/0c:12:a7:54:00:00 Cost = 4 Port = 0x8003
49	31.651588	0.652249	Private_66:68:01	Broadcast	ARP	64	Who has 192.168.33.2 Tell 192.168.33.1
50	31.662899	0.011251	Private_66:68:03	Private_66:68:01	ARP	64	192.168.33.2 Is at 00:50:79:66:68:03
51	31.668249	0.005410	192.168.33.1	192.168.33.2	ICMP	98	Echo (ping) request id=0x7e26, seq=1/256, ttl=64 (reply in 52)
52	31.671957	0.003708	192.168.33.2	192.168.33.1	ICMP	98	Echo (ping) reply id=0x7e26, seq=1/256, ttl=64 (request in 51)
53	32.674154	1.002197	192.168.33.1	192.168.33.2	ICMP	98	Echo (ping) request id=0x7f26, seq=2/512, ttl=64 (reply in 54)
54	32.687692	0.013538	192.168.33.2	192.168.33.1	ICMP	98	Echo (ping) reply id=0x7f26, seq=2/512, ttl=64 (request in 53)
55	33.352831	0.665135	0c:12:a7:54:00:00	192.168.33.2	ICMP	68	Conf. Root = 32768/33/0c:12:a7:54:00:00 Cost = 4 Port = 0x8003
56	33.430241	0.067410	0c:12:a7:54:00:00	192.168.33.2	ICMP	68	Conf. Root = 32768/33/0c:12:a7:54:00:00 Cost = 4 Port = 0x8003
57	33.691501	0.271260	192.168.33.1	192.168.33.2	ICMP	98	Echo (ping) request id=0x8026, seq=3/768, ttl=64 (reply in 58)
58	33.703242	0.011741	192.168.33.2	192.168.33.1	ICMP	98	Echo (ping) reply id=0x8026, seq=3/768, ttl=64 (request in 57)
59	34.709216	1.005974	192.168.33.1	192.168.33.2	ICMP	98	Echo (ping) request id=0x8126, seq=4/1024, ttl=64 (reply in 60)
60	34.722266	0.013050	192.168.33.2	192.168.33.1	ICMP	98	Echo (ping) reply id=0x8126, seq=4/1024, ttl=64 (request in 59)
61	35.724075	1.001809	192.168.33.1	192.168.33.2	ICMP	98	Echo (ping) request id=0x8226, seq=5/1280, ttl=64 (reply in 62)
62	35.742091	0.018016	192.168.33.2	192.168.33.1	ICMP	98	Echo (ping) reply id=0x8226, seq=5/1280, ttl=64 (request in 61)
63	35.766040	0.023949	0c:12:a7:54:00:00	192.168.33.2	ICMP	68	Conf. Root = 32768/33/0c:12:a7:54:00:00 Cost = 4 Port = 0x8003
64	35.830850	0.064810	0c:12:a7:54:00:00	192.168.33.2	ICMP	68	Conf. Root = 32768/33/0c:12:a7:54:00:00 Cost = 4 Port = 0x8003
65	36.283715	0.452865	0c:6a:1b:53:00:00	0c:6a:1b:53:00:00	LOOP	60	Reply

Frame 51: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface v, id 0
Ethernet II, Src: Private_66:68:01 (00:50:79:66:68:01), Dst: Private_66:68:03 (00:50:79:66:68:03)
> Destination: Private_66:68:03 (00:50:79:66:68:03)
> Source: Private_66:68:01 (00:50:79:66:68:01)
Type: IPv4 (0x0800)
[Stream index: 8]
> Internet Protocol Version 4, Src: 192.168.33.1, Dst: 192.168.33.2
> Internet Control Message Protocol

0000 00 50 79 66 68 02 00 50 79 66 68 01 08 00 45 00 Pyth P yfh E
0010 00 54 26 7e 00 00 40 01 90 47 c0 a8 21 01 c0 a8 T6- 8
0020 21 02 08 00 a1 e4 7e 26 00 01 08 03 0a 0b 0c 0d 1-4-
0030 0a 02 10 11 12 13 14 15 16 17 18 19 1a 1b 1c 1d
0040 1e 1f 20 21 22 23 24 25 26 27 28 29 2a 2b 2c 2d ...*8% 4'()*+,-
0050 2e 2f 30 31 32 33 34 35 36 37 38 39 3a 3b 3c 3d ./012345 6789;:;<=
0060 3e 3f >?

Сохранить конфигурации

На каждом коммутаторе:

enable

show running-config